

Exp= 4.1

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to print the second largest integer in a list of arrays

→

include <stdio.h>

int main ()
{

int n, i;

printf (" enter number of elements :");
scanf ("%d", &n);

int arr [n];

printf (" Enter -1. d integers : \n", n);
for (i = 0; i < n; i++)
{scanf ("%d", &arr[i]);
}

int largest, second-largest;

if (arr[0] > arr[1])
{

largest = arr[0];

second-largest = arr[1];

}

else

{

largest = arr[0];

second-largest = arr[1];

}

for (i = 2; i < n; i++)
{

if (arr[i] > largest)
{

second-largest = largest;

largest = arr[i];

}

else if (arr[i] > second-largest && arr[i] < largest)
{

second-largest = arr[i];

}

}


```
printf ("The second largest element is: %d\n",  
        second_largest);
```

```
return 0;
```

```
}
```

exp1.c exp_4.c x C exp2.c exp_4.c C exp3.c exp_4.c C exp1.c exp_5.c C exp2.c exp_5.c C exp3.c exp_5.c

```

p_4.c > C exp1.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i;
6
7      printf("Enter the number of elements: ");
8      scanf("%d", &n);
9
10     int arr[n];
11
12     printf("Enter %d integers:\n", n);
13     for (i = 0; i < n; i++)
14     {
15         scanf("%d", &arr[i]);
16     }
17
18     int largest, second_largest;
19
20     if (arr[0] > arr[1])
21     {
22         largest = arr[0];
23         second_largest = arr[1];
24     }
25     else
26     {
27         largest = arr[1];
28         second_largest = arr[0];
29     }
30
31     for (i = 2; i < n; i++)
32     {
33         if (arr[i] > largest)
34         {
35             second_largest = largest;
36             largest = arr[i];
37         }
38         else if (arr[i] > second_largest && arr[i] < largest)
39         {
40             second_largest = arr[i];
41         }
42     }
43
44     printf("The second largest element is: %d\n", second_largest);
45
46     return 0;
47 }
48

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

C/C++ Compile Run + v [icon] [icon] [icon]

```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\exp1.exe
Enter the number of elements: 5
Enter 5 integers:
1 30 27 42 19
The second largest element is: 30
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output>

```

Exp = 4.2

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to count and display positive, negative, odd and even numbers in an array

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, positive, negative, zero ;
```

```
    positive = 0 ;
```

```
    negative = 0 ;
```

```
    zero = 0 ;
```

```
    printf ("Enter the number of elements : ") ;
```

```
    scanf ("%d", &n) ;
```

```
    int arr[n]
```

```
    printf ("Enter %d elements : ", n) ;
```

```
    for (i = 0 ; i < n ; i++)
```

```
    {
```

```
        scanf ("%d", &n) ;
```

```
    }
```

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```
for (i=0 ; i < n ; i++)
```

```
{
```

```
    if (arr[i] > 0)
```

```
        positive ++;
```

```
    else if (arr[i] < 0)
```

```
        negative ++;
```

```
    else
```

```
        zero ++;
```

```
}
```

```
printf("elements which are positive = %d\n", positive);
```

```
printf("elements which are negative = %d\n", negative);
```

```
printf("elements which are zero = %d\n", zero);
```

```
return 0;
```

```
}
```

exp_4.c > C exp2.c > main()

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, positive , negative , zero ;
6      positive = 0;
7      negative = 0;
8      zero = 0;
9
10     printf("Enter the number of elements: ");
11     scanf("%d", &n);
12
13     int arr[n];
14
15     printf("Enter %d elements: ", n);
16     for (i = 0; i < n; i++)
17     {
18         scanf("%d", &arr[i]);
19     }
20
21     for (i = 0; i < n; i++)
22     {
23         if (arr[i] > 0)
24             positive++;
25         else if (arr[i] < 0)
26             negative++;
27         else
28             zero++;
29     }
30
31     printf("elements which are positive = %d\n" , positive);
32     printf("elements which are negative = %d\n" , negative);
33     printf("elements which are zero = %d\n" , zero);
34
35     return 0;
36 }
```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\exp2.exe

Enter the number of elements: 6

Enter 6 elements: 1 36 0 -8 -52 6

elements which are positive = 3

elements which are negative = 2

elements which are zero = 1

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> █

Exp = 4.3

→

WAP to read a list of integers and store it in a single dimensional array. Write a C program to find the frequency of a particular number in a list of integers

→

#include <stdio.h>

int main ()

{

int n, i, num, count = 0;

printf("Enter the number of elements : ");
scanf("%d", &n);

int arr [n];

printf("Enter 1-d integers : \n", n);
for (i = 0; i < n; i++)
{


```
scanf ("%d", &arr[i]);  
}
```

```
printf ("Enter the number to find its frequency:");  
scanf ("%d", &num);
```

```
for (i=0 ; i < n ; i++)  
{
```

```
    if (arr[i] == num)  
    {
```

```
        count ++;
```

```
    }
```

```
}
```

```
printf ("The frequency of %d is %d\n", num, count);
```

```
return 0;
```

```
}
```

exp_4.c > C exp3.c > main()

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, num, count = 0;
6
7      printf("Enter the number of elements: ");
8      scanf("%d", &n);
9
10     int arr[n];
11
12     printf("Enter %d integers:\n", n);
13     for (i = 0; i < n; i++)
14     {
15         scanf("%d", &arr[i]);
16     }
17
18     printf("Enter the number to find its frequency: ");
19     scanf("%d", &num);
20
21     for (i = 0; i < n; i++)
22     {
23         if (arr[i] == num)
24         {
25             count++;
26         }
27     }
28
29     printf("The frequency of %d is: %d\n", num, count);
30
31     return 0;
32 }
33
```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output'

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> & .\exp3.exe

Enter the number of elements: 10

Enter 10 integers:

1 76 0 -23 0 11 1 64 0 -78

Enter the number to find its frequency: 1

The frequency of 1 is: 2

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_4.c\output> |

Exp = 4.4

→

WAP that reads two matrices and computes product. Print both input matrices and resultant matrix with suitable headings & output should be in matrix format only. Program must check the compatibility of orders of matrices for multiplication. Report appropriate message in case of incompatibility

→

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, p, q, r, j, k;
```

```
    printf("Enter rows & columns of matrix A:");
```

```
    scanf("%d %d", &m, &n);
```

```
    int A[m][n];
```

```
    printf("Enter elements of matrix A (row-wise) = m")
```

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```
for (i = 0; i < m; i++)  
{  
    for (j = 0; j < n; j++)  
    {  
        scanf ("%d", &A[i][j]);  
    }  
}
```

```
printf ("Enter rows & columns of matrix B: ");  
scanf ("%d %d", &p, &q);
```

```
int B [p][q];
```

```
printf ("Enter elements of matrix B (row wise) : \n");
```

```
for (i = 0; i < p; i++)  
{
```

```
    for (j = 0; j < q; j++)  
    {
```

```
        scanf ("%d", &B[i][j]);
```

```
    }
```

```
}
```

```
if (n != p)
```

```
    printf ("\n Matrix multiplication not possible \n");
```

```
    printf ("Reason: columns of A (%d) not equal to
```

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```
Row of (%d)\n", n, p);  
return 0;  
}
```

```
int C[m][q]  
{  
    for (i=0; i<m; i++)  
    {  
        for (j=0; j<q; j++)  
        {  
            C[i][j] = 0;  
            for (k=0; k<n; k++)  
            {  
                C[i][j] = C[i][j] + A[i][k] * B[k][j];  
            }  
        }  
    }  
}
```

```
printf ("In matrix A: \n");  
for (i=0; i<m; i++)  
{  
    for (j=0; j<n; j++)  
    {  
        printf ("%d", A[i][j]);  
    }  
    printf ("\n");  
}
```

```
printf ("In matrix B : \n" );  
for ( i = 0 ; i < p ; i++)  
{  
    for ( j = 0 ; j < q ; j++)  
    {  
        printf ("%4d", B[i][j]) ;  
    }  
    printf (" \n " ) ;  
}
```

```
printf ("In Resultant Matrix (A x B) : \n" );  
for ( i = 0 ; i < m ; i++)  
{  
    for ( j = 0 ; j < q ; j++)  
    {  
        printf ("%4d", C[i][j]) ;  
    }  
    printf (" \n " ) ;  
}
```

```
return 0 ;
```

```
}
```



```

exp_4.c > C exp4.c @ main()
1 #include <stdio.h>
2
3 int main()
4 {
5     int m, n, p, q, i, j, k;
6
7     printf("Enter rows and columns of Matrix A: ");
8     scanf("%d %d", &m, &n);
9
10    int A[m][n];
11    printf("Enter elements of Matrix A (row-wise):\n");
12    for (i = 0; i < m; i++)
13    {
14        for (j = 0; j < n; j++)
15        {
16            scanf("%d", &A[i][j]);
17        }
18    }
19
20    printf("Enter rows and columns of Matrix B: ");
21    scanf("%d %d", &p, &q);
22
23    int B[p][q];
24    printf("Enter elements of Matrix B (row-wise):\n");
25    for (i = 0; i < p; i++)
26    {
27        for (j = 0; j < q; j++)
28        {
29            scanf("%d", &B[i][j]);
30        }
31    }
32
33    if (n != p)
34    {
35        printf("\nMatrix multiplication not possible!\n");
36        printf("Reason: Columns of A (%d) not equal to Rows of B (%d)\n", n, p);
37        return 0;
38    }
39
40    int C[m][q];
41    for (i = 0; i < m; i++)
42    {
43        for (j = 0; j < q; j++)
44        {
45            C[i][j] = 0;
46            for (k = 0; k < n; k++)
47            {
48                C[i][j] = C[i][j] + A[i][k] * B[k][j];
49            }
50        }
51    }
52
53    printf("\nMatrix A:\n");
54    for (i = 0; i < m; i++)
55    {
56        for (j = 0; j < n; j++)
57        {
58            printf("%d", A[i][j]);
59        }
60        printf("\n");
61    }
62
63    printf("\nMatrix B:\n");
64    for (i = 0; i < p; i++)
65    {
66        for (j = 0; j < q; j++)
67        {
68            printf("%d", B[i][j]);
69        }
70        printf("\n");
71    }
72
73    printf("\nResultant Matrix (A x B):\n");
74    for (i = 0; i < m; i++)
75    {
76        for (j = 0; j < q; j++)
77        {
78            printf("%d", C[i][j]);
79        }
80        printf("\n");
81    }
82
83    return 0;
84 }
85

```

```

P5 C:\Users\sanchi\OneDrive\Desktop\c programming> cd "c:\Users\sanchi\OneDrive\Desktop\c programming\exp_4.c\output"
P5 C:\Users\sanchi\OneDrive\Desktop\c programming\exp_4.c\output> & .\exp4.exe
Enter rows and columns of Matrix A: 2 3
Enter elements of Matrix A (row-wise):
1 2 3 4 5 6
Enter rows and columns of Matrix B: 3 2
Enter elements of Matrix B (row-wise):
7 8 9 10 11 12

Matrix A:
1 2 3
4 5 6

Matrix B:
7 8
9 10
11 12

Resultant Matrix (A x B):
58 64
138 154
P5 C:\Users\sanchi\OneDrive\Desktop\c programming\exp_4.c\output>

```